DualBalance Districting:
From Detection to Generation

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Abstract

Most current work on gerrymandering tries to detect it: identify when a line-drawer has crossed from neutral redistricting into partisan or racial manipulation. We generate maps instead. *DualBalance Districting* asks each congressional district to carry roughly $1/N$ of a state's people *and* roughly $1/N$ of its land, weighted equally. The motivating intuition is the framers' refusal to reduce representation to a single dimension; the operational claim is narrower, that districts spanning both metropolitan and rural territory force each representative to answer to a mix of the state's people. *DualBalance* places N seeds in a small circle around the state's population-weighted centroid and assigns each census unit to the nearest seed with capacity remaining. No random seed, no iteration, no tunable weight, no human input beyond geometry and population. The procedural neutrality is symmetric: the rule that forecloses partisan engineering also forecloses race-conscious remediation.

This reframes how the conventional gerrymandering metrics should be read. Compactness scores, the efficiency gap, mean-median asymmetry, ensemble outlier tests, and majority-minority counts play two roles on enacted plans: they describe plan effects, and they support inferences about the line-drawer's intent. The descriptive role survives on a DualBalance map; the intent-attribution role does not.

Validated against enacted 119^{th}-Congress plans across all available states using 2020 PL 94-171 data, DualBalance beats the enacted plan on the DualBalance Score in the large majority of states and achieves *Karcher*-compliant population balance on nearly all of them. We report standard partisan and race diagnostics alongside, as descriptions of the partition.

1 Introduction

In most U.S. states, congressional district boundaries are drawn by the same legislators whose electoral prospects depend on where those boundaries fall. This structural conflict of interest is not resolved by the Constitution, which mandates only that House districts achieve population balance (Art. I, §2) while leaving every other line-drawing decision to political actors. The consequences are well-documented: gerrymandered boundaries reduce electoral competition, dilute minority voting power, and produce seat allocations that diverge substantially from statewide vote totals [2, 15]. Federal judicial oversight of these practices has progressively narrowed, as detailed in §1.1 below. This paper proposes and validates *DualBalance Districting*: a deterministic algorithm that computes congressional district maps as a pure function of state geometry, census populations, and district count N, with no free parameters, no randomness, and no human intervention at any stage. Each district is designed to carry approximately $1/N$ of the state's population and approximately $1/N$ of its land area, weighted equally in a single objective we call the DualBalance Score. A deterministic algorithm with no free parameters provides, by construction, a mathematical barrier against intentional map manipulation.

1.1 Redistricting Without a Neutral Standard

The federal courts have largely withdrawn from gerrymandering review. In *Rucho v. Common Cause* [21], the Supreme Court held 5–4 that partisan-gerrymandering claims present nonjusticiable political questions; the Court conceded the practice is "incompatible with democratic principles" but found no manageable judicial standard. After *Rucho*, the only federal channel for gerrymandering review is racial. That channel has since narrowed. Section 2 of the Voting Rights Act, as construed in *Thornburg v. Gingles* [19], requires majority-minority districts where the three Gingles preconditions are met. The Court reaffirmed this in *Allen v. Milligan* [22]. One year later, *Alexander v. South Carolina Conf. of the NAACP* [24] raised plaintiffs' evidentiary burden; *Louisiana v. Callais* [25] held in April 2026 that even a plan drawn to comply with a prior Section 2 order may violate the Equal Protection Clause if Section 2 did not in fact compel a race-based remedy. Together, *Alexander* and *Callais* leave race-conscious line-drawing in a narrower window of constitutional safety than at any point since the VRA's adoption.

State constitutional review under *Moore v. Harper* [23] remains available but is uneven and politically contingent: roughly ten state supreme courts have recognized state-constitutional partisan-gerrymandering claims [11]; the rest have not. Several states have already redrawn maps mid-decade in response to electoral results rather than census revision, shifting redistricting from a once-per-decade event into a recurring one. Reducing this discretionary intervention requires a procedure whose output is determined entirely by census data, with no human choices at the line-drawing stage.

1.2 Population and Geography as Dual Representational Axes

The Constitution already encodes two theories of representation: the House apportions seats by population (Art. I, §2), the Senate by sovereign state regardless of population (Art. I, §3). Madison's defense in *Federalist* Nos. 54–58 [8] frames the bicameral structure as a principled refusal to collapse representation onto a single dimension. Within the House, the existing framework uses only one extensive quantity, population, to define what a district should be. Geographic area is a second extensive quantity: it is well-defined, measurable from the same census geometry, and orthogonal to population density. Balancing area alongside population discourages concentration of districts exclusively within either dense metropolitan cores or sparse rural regions.

The Electoral College makes the same dual logic operative at the executive level. Each state's electoral vote total equals its House seats plus its two Senate seats (Art. II, §1), explicitly combining a population-proportional component with a geographic floor. Presidential candidates must assemble coalitions distributed across states, not merely population-weighted majorities. Congressional districts calibrated on population alone are, in this respect, the exception in the broader constitutional framework rather than the rule.

This is a mathematical challenge, not a free choice. A typical U.S. state varies by two to three orders of magnitude between its densest census tract and its sparsest rural block. Equal population, equal area, contiguity, and shape compactness cannot all be satisfied simultaneously in general; some trade-off is unavoidable. The question is which trade-off to make, how transparently to make it, and whether the resulting procedure can be defended as impartial.

1.3 Existing Metrics Are Forensic, Not Generative

Quantitative tests for gerrymandering include shape-compactness measures (Polsby-Popper [12]; Reock [13]), partisan-asymmetry statistics [30, 31], the Efficiency Gap [15], and ensemble outlier methods [4, 5]. We report Polsby-Popper, Reock, and the Efficiency Gap (EG) as comparative benchmarks in §3. EG, defined formally in §2, measures the difference between the two parties' wasted-vote totals as a fraction of statewide turnout; it has received the widest adoption in federal redistricting litigation, serving as the central metric in *Whitford v. Gill*, with $|\text{EG}| > 0.07$ proposed as a threshold for a plausible partisan-gerrymandering challenge [15]. All of these metrics are *forensic* instruments: they take an enacted map as input and ask whether its observed properties (shape, vote efficiency, partisan asymmetry, the location of its seat–vote curve) are consistent with maps that some reference process (random redistricting, ensemble sampling under legal constraints) would have produced absent partisan intent. They compare the enacted plan against a counterfactual distribution to infer the line-drawer's motive, and were built for an adversarial context in which the question is "did somebody do something here that they should not have done?"

A generative metric is a different object. It states, directly, what a district *should* look like, and is used as the objective of the line-drawing procedure rather than as evidence about the procedure. Population balance is the only generative metric U.S. redistricting law currently uses: *Wesberry v. Sanders* [17] and *Reynolds v. Sims* [16] require that each district hold roughly $1/N$ of the relevant population. Every other criterion (compactness, communities of interest, partisan fairness) enters the law as a forensic instrument or a discretionary guideline.

The DualBalance Score (DBS; equation 1 in §2) is a *generative* criterion: it states what a district should be, one that carries roughly $1/N$ of the people and roughly $1/N$ of the land, and serves as the objective the algorithm pursues rather than a test applied after the fact.

1.4 The Iowa Model as a Reference Point

Iowa's Legislative Services Agency [6] has operated under a written, publicly documented redistricting procedure since 1980 – one of the few U.S. examples of a formalized, procedural approach that has functioned without major legal challenge across multiple redistricting cycles. Its longevity and transparency make it a natural reference point for proposals to replace discretionary redistricting with an explicit algorithm. The open question is whether a procedure designed for Iowa's geographic and demographic conditions constitutes a legally defensible, impartial template for states with substantially different population distributions and geographic structure. We re-implement it as a structured baseline (referred to here as *Cascade* to avoid confusion with the agency) and use it as a point of comparison throughout.

1.5 Contribution

We propose *DualBalance Districting*: a deterministic multi-resolution pipeline whose output is a pure function of (state geometry, census populations, N), formalized in §2.

Pipeline. The algorithm runs in three stages: radial seed placement around the population-weighted centroid, followed by a two-phase greedy local search at VTD scale that closes per-district population deviation toward the *Karcher* threshold [18], followed by the same search re-run at census-block scale where finer granularity allows the DualBalance Score to make its largest gains. The full formalization is in §2.

Empirical validation. We run the full pipeline on six states selected to cover the geographic range that breaks Cascade: IA, MA, MN, NC, TX, WI. Five of six achieve a DualBalance Score above the enacted 119th-Congress plan. Two (IA, WI) reach the 0.05% Karcher threshold exactly; three more (MN, NC, MA) sit within a factor of two of it; TX, the hardest case, sits at 0.52%, five times tighter than the same state's enacted plan. On partisan fairness, DualBalance's Efficiency Gap is smaller in magnitude than the enacted plan's in four of six states, with the largest improvements on the three most heavily-litigated maps (NC, WI, TX). On minority-majority district counts, DualBalance produces more such districts than the enacted plan on NC (2 vs. 1) and TX (22 vs. 19).

Claims. The procedure contains no explicit partisan, racial, or incumbent-aware optimization criteria: its inputs are census geometry and population totals, and its output is a pure function of those inputs with no tunable parameters affecting electoral or demographic outcomes. Adopting a generative criterion as the design objective, rather than a forensic instrument, is a natural response to a setting in which no line-drawer is present to investigate. Scope limits and further qualifications appear in §4.

Roadmap. Section 2 formalizes the algorithm. Section 3 reports the six-state empirical validation. Section 4 discusses limits, trade-offs, and legal exposure.

2 Methods

2.1 Data and inputs

Atomic units are U.S. Census Bureau *voting tabulation districts* (VTDs) and *census tabulation blocks* from the 2020 decennial release [28], obtained via TIGER/Line shapefiles [27]. Boundaries are used as-is; no simplification or smoothing is applied. Population counts are the total-population figures from PL 94-171. The enacted congressional plan used for comparison is the 119th-Congress plan (effective January 2025), joined to VTDs and blocks via a spatial representative-point join against the TIGER 2024 `cd119` layer [29].

2.2 Problem statement

Given a state S partitioned into atomic units $U = \{u_1, \dots, u_M\}$ (census tabulation blocks, block groups, or voting tabulation districts (VTDs), in increasing order of size) and a target district count N, DualBalance Districting produces a deterministic assignment $\pi : U \to \{1, \dots, N\}$ such that each district $D_i = \pi^{-1}(i)$ is contiguous, non-empty, and as close as the geometry allows to representing both $1/N$ of the state's people and $1/N$ of its geography. Let $P = \sum_u \mathrm{pop}(u)$ and $A = \sum_u \mathrm{area}(u)$ with per-district targets $P^* = P/N$ and $A^* = A/N$.

The full pipeline runs in three stages: a radial-seed stage, *DualBalance* (§§2.3–2.5), which provides the deterministic starting configuration; a VTD-scale tightening stage (§2.7) which drives $\max_i |\delta_i|/P^*$ toward the user-supplied tolerance τ via a hybrid greedy local search with bounded-chain escape; and a block-scale refinement stage (§2.8) which re-initializes at finer granularity to recover area balance within the achieved pop-deviation envelope. Each stage is a pure function of its inputs.

2.3 Radial seed placement

Compute the population-weighted centroid

$$c = (c_x, c_y) = \left(\frac{\sum_u p_u\, x_u}{\sum_u p_u}, \frac{\sum_u p_u\, y_u}{\sum_u p_u} \right),$$

where x_u, y_u are the centroid coordinates of unit u in an equal-area projection and p_u its population. Let diag denote the bounding-box diagonal of the units and set the seed radius $r = 0.001 \cdot \text{diag}$. For $d = 0, 1, \ldots, N-1$ place seed s_d at

$$s_d = \big(c_x + r\cos(2\pi d/N),\ c_y + r\sin(2\pi d/N)\big).$$

Seed 0 sits due east of the centroid; seeds advance counter-clockwise by equal angular steps. The choice $r = 0.001 \cdot \text{diag}$ is small enough that the Voronoi cells generated by the seeds degenerate to near-perfect radial slices through c, but large enough to keep the seed positions numerically distinct.

The radial configuration is what carries the dual-balance property: each slice spans both the dense (near-c) and sparse (boundary-side) territory of the state, so the population it inherits is bounded by the cap (§2.4) while the area it inherits is driven toward A^* by the slicing geometry.

2.4 Capacitated first-fit assignment

Let $d(u, i) = \|x_u - s_i\|$ be the Euclidean distance from unit u to seed i. Initialize remaining capacities $\rho_i = P^*$ for $i = 1, \ldots, N$. Sort all (u, i) pairs by ascending normalized distance $d(u, i)/\text{diag}$ and walk the sorted list in order:

for each (u, i) in ascending $d(u, i)/\text{diag}$:
  if u already assigned: continue
  if $\rho_i \geq p_u$: assign u to i; $\rho_i \leftarrow \rho_i - p_u$
  else: skip

Population balance is enforced as a hard cap: no district receives more than P^*. Any unit not placed by the end of the walk (a rare integer-rounding edge case) is assigned to the district with the largest remaining capacity; `argmax` resolves ties to the smallest district id.

Ties in normalized distance break by ascending (unit_id, district_id). There is no Lloyd recentering, no iteration count, no convergence test: the radial seeds do not drift, so a single assignment pass suffices.

2.5 Contiguity repair

After §2.4 every unit belongs to exactly one district, but a district may consist of more than one connected component (rare on convex states; more common on those with peninsulas, islands, or rural enclaves). For each such district, the largest connected component by total population is retained; units in the smaller components are reassigned to neighboring districts one at a time, in the lowest-cost direction. Cost is

$$c(u, j) = \frac{d(x_u, s_j)}{\text{diag}} + \frac{|P(D_j) + p_u - P^*|}{P^*} + \frac{|A(D_j) + a_u - A^*|}{A^*},$$

combining a normalized distance term with normalized population and area deviation penalties for the receiving district. Ties break in cascade $(c, \text{pop\_pen}, \text{area\_pen}, \text{dist}, \text{district\_id})$ ascending. The repair sweep iterates until no district has more than one connected component, capped at ten sweeps; in practice it converges in zero or one sweep on real census geometries.

2.6 Scoring

Define per-district relative deviations $\text{pop\_dev}_i = |P(D_i) - P^*|/P^*$ and $\text{area\_dev}_i = |A(D_i) - A^*|/A^*$ and let $\overline{\text{pop\_dev}}$, $\overline{\text{area\_dev}}$ denote their means over $i = 1, \ldots, N$. The DualBalance Score is

$$\text{DBS} = \frac{1}{1 + \frac{1}{2}\,\overline{\text{pop\_dev}} + \frac{1}{2}\,\overline{\text{area\_dev}}}. \tag{1}$$

The 0.5/0.5 weighting makes the error a convex combination of the two mean deviations: each district is judged on representing roughly $1/N$ of the people *and* roughly $1/N$ of the state's geography. The score reaches 1.0 for a perfectly balanced plan and approaches 0 as deviations grow without bound. Secondary metrics (Polsby-Popper compactness [12] and Reock [13]) are computed alongside but not optimized against; the Phase 2 optimizer (§2.7) hill-climbs DBS, not the compactness metrics. Radial slices have lower compactness than blob-Voronoi or hand-drawn districts by construction; this is a deliberate trade in service of the dual-balance objective.

DualBalance does not directly minimize (1); it minimizes population-capacitated geographic assignment cost under radial seeding. The empirical consequences are reported in §3.

For partisan-fairness comparison we report the Efficiency Gap (EG) [15]. Let $\text{wasted}_P(d)$ denote the wasted votes for party P in district d: all votes cast for a losing candidate, plus votes above the bare majority threshold for a winning candidate. Then

$$\text{EG} = \frac{\sum_d \text{wasted}_R(d) - \sum_d \text{wasted}_D(d)}{\sum_d \text{total}(d)}, \tag{2}$$

where positive values indicate Republican advantage and negative values indicate Democratic advantage. EG is computed on the plan geometry using precinct-level presidential vote totals from the 2020 election; it is reported as a diagnostic and does not enter the generator. All other partisan, racial, and demographic variables are similarly excluded from the generator's inputs.

2.7 $\mathbf{L^1}$ population tightening and DBS hill-climb

The radial pipeline produces per-district pop deviation in the 5–15% range on real census geometry, well above the ˜0.5% threshold required by *Reynolds v. Sims* for U.S. congressional districts [16]. A deterministic two-phase local search of boundary-unit moves (activated via `--tighten-pop`) closes this gap; all results reported in this paper use this pass. Let $\delta_i = P(D_i) - P^*$ denote the signed population deviation of district D_i, let T denote the user-supplied tolerance (default 0.005), and call a move *safe* if the source unit is not an articulation point of its district's induced subgraph (so that removing it preserves contiguity).

Phase 1: population tightening (hybrid $\mathbf{L^1}$ + max-norm). At each step, scan every boundary unit and every neighboring district and compute both the L^1 change

$$\Delta_{L^1}(u, d_{\text{dest}}) = |\delta_{d_{\text{src}}} - p_u| - |\delta_{d_{\text{src}}}| + |\delta_{d_{\text{dest}}} + p_u| - |\delta_{d_{\text{dest}}}|$$

and the max-norm effect: a move *strictly reduces* the global maximum iff $|\delta_{d_{\text{src}}} - p_u| < \max_i |\delta_i|$, $|\delta_{d_{\text{dest}}} + p_u| < \max_i |\delta_i|$, and at least one of $\delta_{d_{\text{src}}}, \delta_{d_{\text{dest}}}$ is itself at the current maximum. Each scanned candidate is labeled with a priority key: priority 0 if it strictly reduces the max, priority 1 if it only reduces L^1. Candidates are ranked lexicographically by $(\text{priority}, \Delta_{L^1})$, and the safe top-ranked candidate is accepted.

This hybrid is essential. Pure L^1-greedy leaves the worst district untouched when it sits between two other over-target districts: draining it requires L^1-neutral moves that the L^1 criterion rejects.

Pure max-norm gets stuck symmetrically: when two districts are tied at the max, no single move strictly reduces the global maximum (draining one leaves the other at the same value). The hybrid takes any move that helps either norm, with max-reducing moves preferred so the worst district is targeted whenever possible.

When neither 1-opt criterion finds an improving move but $\max_i |\delta_i|/P^* > T$, invoke a bounded *chain escape*: search for an augmenting transport on the district-adjacency graph of the form

$$u_0 : D_0 \to D_1,\ u_1 : D_1 \to D_2,\ \ldots,\ u_{k-1} : D_{k-1} \to D_k,$$

of length $k = 2$ then $k = 3$, in which each u_j is a boundary unit between D_j and D_{j+1}. The chain is accepted if it strictly cuts the global maximum, or if it holds the maximum flat while strictly reducing the L^1 sum (which is what happens when several districts are tied at the max: each chain drains one of them, the global maximum stays at the same value until the last tied district is drained, but the L^1 sum drops monotonically). The unit-level search inside each district triple sorts the arc lists $B_{i\to j}$ by population value; matching pairs (u, v) are found via binary search around a target population value rather than by Cartesian enumeration, bounding the per-triple cost by $O(|B_{i\to j}| \log |B_{j\to k}|)$. Application is in *reverse* order (u_{k-1} first, then u_{k-2}, etc.) so the intermediate districts never temporarily overload during the chain. This is the deterministic analogue of an ejection chain. Phase 1 terminates only when neither 1-opt nor bounded-chain escape finds an improving sequence.

The L^1 objective is essential to the radial geometry. Seed placement puts the most over-target and most under-target districts on opposite sides of the population centroid, so no single adjacent-slice swap reduces the L^∞ maximum, but many such swaps reduce the sum. A pure max-norm criterion stalls on this geometry; the hybrid with chain escape continues to make progress where the L^∞-only criterion would halt.

Phase 2: DBS hill-climb. Once Phase 1 has converged, at each step pick the safe boundary move that maximally improves DBS (equation 1), subject to $\max_i |\delta_i|/P^*$ not exceeding the value Phase 1 reached (equivalently, never crossing back above T). Phase 2 runs until no improving safe move exists. Whereas the core DualBalance pass does not directly minimize equation 1, this opt-in Phase 2 does, but only on the post-Phase-1 plan and only within the Reynolds-compliant feasible region.

Determinism and engineering. Both phases are pure functions of (units, N, T). The only source of dependency beyond the core pipeline's (units, N) is the explicit tolerance T. The optimizer maintains a per-district articulation-point cache via Tarjan's algorithm on CSR adjacency arrays [26], reducing the per-candidate contiguity check from $O(V + E)$ to $O(1)$; an incrementally tracked boundary-unit set restricts each scan to units that have a different-district neighbor. At VTD scale the speedup is modest; at block scale (tens of thousands of units per district) it is the difference between hours and minutes.

The pass is off by default and gated by an explicit flag. Pure radial remains the principled algorithm; the two-phase optimizer is an opt-in concession to legal compliance, with Phase 2 providing the further courtesy of directly hill-climbing the project's stated objective once the Reynolds constraint is satisfied.

2.8 Multi-resolution refinement: VTD to Census block

The hybrid Phase 1 + chain escape closes most of the Reynolds gap at VTD scale, but not all of it. Modern U.S. congressional plans must clear the much tighter *Karcher v. Daggett* [18] practical threshold of $\sim 0.05\%$ total deviation, and at VTD granularity the residual gap is structural rather than algorithmic. The smallest single-unit move has the size of the smallest VTD's population. On real states the median VTD population is on the order of 10^3, while the Karcher budget on a ~ 800k ideal district is ~ 400 people. Most candidate moves overshoot the budget on at least one endpoint, and Phase 2 starves: the running-max constraint admits few moves, and the algorithm settles above Karcher with most of the residual area-balance gain unrealized.

The fix is to refine at finer granularity. We re-run the optimizer on Census tabulation blocks (median population ~ 20 rather than $\sim 10^3$), initializing from the VTD-Karcher plan rather than re-seeding from scratch. The pipeline is:

1. Compute the VTD-level plan $\pi_{\text{VTD}} : V_{\text{VTD}} \to \{1, \ldots, N\}$ via DualBalance + Phase 1 + Phase 2 at VTD scale with tolerance T.
2. Load the block-level units V_{block} and project both layers to the same equal-area CRS.
3. For each block $u \in V_{\text{block}}$, find the VTD polygon $v(u) \in V_{\text{VTD}}$ containing its representative point, and set $\pi_0(u) := \pi_{\text{VTD}}(v(u))$. Blocks whose representative point lies outside every VTD polygon (rare, only on coastal/water boundary cases) fall back to the nearest VTD by centroid distance. The spatial join is deterministic given a fixed tie-breaking order on unit_id.
4. Run Phase 1 + Phase 2 again at block scale starting from π_0, with the same tolerance T.

Because VTDs are unions of whole blocks, π_0 inherits the VTD-level plan's pop totals and contiguity exactly: the block-level plan is identical to the VTD-level plan re-described at finer granularity. Phase 1 at block scale therefore has nothing to do on states where the VTD-level pass reached T. Phase 2 has substantial freedom: block populations are two orders of magnitude smaller than VTD populations, so the running-max envelope around T admits many candidate moves and the DBS hill-climb runs until saturation.

Because DualBalance-style seeding at block scale starts from a much worse initial configuration than the VTD seed (the population-weighted centroid is the same but the much higher density of degenerate articulation-point boundaries traps the optimizer), the refinement strategy is the path that works in practice: VTD as a coarse solver, blocks for the precise area-balance recovery. Contiguity checks within the optimizer use a per-district articulation-point cache (Tarjan's algorithm [26] on CSR adjacency arrays), reducing each query from $\mathcal{O}(|V_d| + |E_d|)$ to $\mathcal{O}(1)$ amortized; this is what makes the block-scale stage tractable on commodity hardware.

3 Results

We evaluated DualBalance against the enacted 119th-Congress plan on all multi-seat states for which TIGER 2020PL VTD boundaries are available. California, Hawaii, and Oregon did not submit VTD data to the Census Bureau and are excluded throughout (marked † in figures and tables). The same data pipeline runs unmodified for every state via `scripts/prep_state_units.py`: TIGER 2020 VTDs, Census PL 94-171 demographics, dra2020/vtd_data 2020 presidential returns, and a TIGER 2024 cd119 spatial join for the enacted plan. To anchor DualBalance in context we score two additional deterministic algorithms: *Cascade* (`src/dualbalance/cascade.py`), an Iowa-LSA-flavored construction that aggregates VTDs to counties and uses farthest-point seeding; and *BDistricting* [9], Brian Olson's published 50-state map series ingested via Census 2020 Block Assignment Files (`scripts/prep_bdistricting.py`). All outputs are byte-identical across repeated runs on the same input; the CLI integration test `test_generate_determinism_via_cli` pins this in CI.

Three cross-state findings organize the presentation.

Partisan asymmetry shrinks under every deterministic generator. On states with nonzero enacted efficiency gap, all three deterministic algorithms produce smaller |EG| than the enacted plan (Figure 1). The largest reductions appear in states most associated with partisan-gerrymander litigation. The structural explanation is in §4.7: a generator that reads no political data cannot reproduce a large partisan-fairness asymmetry. This holds across all three deterministic baselines, not only DualBalance.

Cascade wins on DBS but is legally non-viable on most states. Cascade scores well on the DualBalance objective because county aggregation naturally produces units of similar area, but the county-integrity constraint yields population deviations far above the *Karcher* threshold on any state with a dominant metropolitan county (Figure 2, Panel B). DualBalance and BDistricting both achieve *Karcher* compliance on the large majority of states; Cascade does so only on Iowa and Wisconsin, where no single county exceeds the per-district population cap. A plan that wins on DBS but violates *Reynolds v. Sims* cannot legally be enacted.

Different deterministic algorithms, different trade-offs. DualBalance holds pop_dev_max below 1 % on nearly all states and recovers area balance through radial mixing of dense and sparse territory, at the cost of lower Polsby-Popper compactness (Figure 2, Panels A and D). BDistricting maximizes compactness and population balance but accepts high area imbalance because Lloyd recentering explicitly minimizes within-district geographic spread. Cascade maximizes county integrity and area balance but fails the population-balance legal threshold on most states.

[PLACEHOLDER — Table 1]

One row per state (50 rows). Columns: State, N, DualBalance Score (DualBalance / Cascade / BDistricting / Enacted), pop_dev_max (DualBalance / Cascade). Bold marks the DBS winner per row. Cascade rows flagged ‡ where pop_dev_max > 0.5 % (legally non-viable). States marked † excluded (no TIGER VTD data).

Source: `out/compare_all_summary.json`.

Table 1: Per-state DualBalance results for all 50 states. † State excluded (TIGER 2020PL VTD data not available). ‡ pop_dev_max > 0.5 %; plan does not satisfy *Reynolds v. Sims* and is legally non-viable regardless of DBS. **Bold** marks the highest DBS on each row.

3.1 What would Congress look like?

[PLACEHOLDER, ∼150 words.] Aggregate seats_R / seats_D summed across all available states under DualBalance, Cascade, BDistricting, and the enacted plan, plus a proportional-vote baseline derived from the 2020 statewide two-party presidential returns. The question, “if every

[PLACEHOLDER — Table 2]

One row per metric, one column per algorithm (DualBalance / Cascade / BDistricting / Enacted). Rows: DBS median, pop_dev_max median, states at *Karcher* ($\leq 0.05\,\%$), states beating enacted DBS, |EG| median, Polsby-Popper median. Bold marks the best value per row.

Source: `out/compare_all_summary.json`.

Table 2: Aggregate algorithm comparison across all available states. **Bold** marks the best value in each row. "At *Karcher*" counts states where pop_dev_max $\leq 0.05\,\%$. "Beats enacted DBS" counts states where the algorithm's DualBalance Score exceeds the enacted plan's.

[PLACEHOLDER — Figure 1]

Ranked bar chart: |EG| for DualBalance vs. enacted 119th-Congress plan, one bar pair per state, sorted left-to-right by enacted |EG| (worst-gerrymandered on the left). Red dashed line at $|\text{EG}| = 0.07$ (Stephanopoulos-McGhee threshold). Source: `out/story_headline_eg.png`. States without TIGER 2020PL VTD data excluded (†).

Figure 1: Partisan fairness across all available states, sorted by enacted |EG| (worst-gerrymandered on the left). DualBalance (blue) vs. enacted 119th-Congress plan (gray). Red dashed line: $|\text{EG}| = 0.07$ gerrymander threshold [15]. States marked † lacked TIGER 2020PL VTD boundaries and are excluded.

[PLACEHOLDER — Figure 2]

2×2 boxplot figure. Each panel: one box per algorithm (DualBalance, Cascade, BDistricting, Enacted), one dot per state.

Panel A: DualBalance Score (higher = better).
Panel B: pop_dev_max, log scale; red dashed line at *Karcher* 0.05 %.
Panel C: |EG| (lower = fairer); red dashed line at 0.07.
Panel D: Polsby-Popper mean (the honest compactness admission).

Source: `out/story_box_*.png`.

Figure 2: Cross-state comparison of four algorithms on key metrics. Each box spans the interquartile range across all available states; dots are individual states. **Panel A**: DualBalance Score (higher = better). **Panel B**: maximum per-district population deviation, log scale; dashed line at the *Karcher* threshold (0.05 %). **Panel C**: |EG| (lower = fairer); dashed line at the 0.07 gerrymander threshold [15]. **Panel D**: Polsby-Popper compactness (mean per state); DualBalance is structurally less compact than enacted plans because radial slices are not blob-shaped.

[PLACEHOLDER — Figure 3]

Scatter plot: minority-majority district count under DualBalance (y-axis) vs. enacted plan (x-axis), one point per state, diagonal reference line $y = x$. Points above the diagonal: DualBalance creates more MMDs than enacted; below: fewer; on the line: tied. Notable labeled points: NC (1→2), TX (19→22) above the diagonal; MA (1→0), WI (1→0) below. Most states cluster at $(0, 0)$.

Source: computed from `out/compare_all_summary.json`.

Figure 3: Minority-majority district count: DualBalance vs. enacted 119th-Congress plan. Each point is one state; the diagonal is $y = x$. Points above the line indicate DualBalance produces more majority-minority districts than the enacted map; points below indicate fewer. Most states cluster at $(0, 0)$ because no single minority group concentrates densely enough to support a majority-minority district of standard size. DualBalance is race-blind; where it produces more majority-minority districts than the enacted plan, the effect is geographic, not by design. Where it produces fewer (notably MA and WI), the enacted plan constructed a majority-minority district that radial slicing breaks up.

[PLACEHOLDER — Figure 4]

Three-panel side-by-side map: North Carolina (14 districts).
Left: Enacted 119th-Congress plan (EG +0.20, pop_dev_max 0.04 %).
Center: Cascade (EG +0.03, pop_dev_max 10.27 %, legally non-viable).
Right: DualBalance (EG +0.09, pop_dev_max 0.11 %).
Source: `out/maps_NC.png`.

Figure 4: North Carolina congressional districts under three plans (14 seats, 2020 PL 94-171). **Left:** enacted 119th-Congress plan, with the Efficiency Gap of +0.20 that gave rise to *Rucho v. Common Cause* [21]. **Center:** Cascade plan, which scores better on DBS (0.82 vs. 0.77 enacted) but violates *Karcher* at 10.27 % maximum population deviation and cannot legally be enacted. **Right:** DualBalance plan, *Karcher*-compliant (0.11 %) and reducing EG to +0.09 with no political input.

state used DualBalance, what is the partisan composition of the House?", is answered here with full caveats: 2020 presidential returns proxy for House votes, CA/HI/OR are excluded for lack of VTD data, and single-seat states are unchanged under any algorithm. The full per-state breakdown (seats R/D, statewide R share, all four algorithms) appears in Supplementary Table S1.

4 Discussion

4.1 What the Minnesota result does and does not show

The headline (DualBalance beats the enacted Minnesota plan on the DualBalance Score) is a single-state existence result. It shows that radial slicing can score competitively on the dual-balance objective without reading party, race, or any discretionary input. It does not show that DualBalance generalizes to all 50 states, that DBS is the right scalar to optimize, or that DualBalance maps would survive judicial scrutiny. We take those up below.

4.2 Co-design of score and algorithm

A natural objection: "A beats B on Metric M because A and M were co-designed." DBS (1) was specified before the algorithm choice. We tested three forms: the weighted form used here, a sum-of-means form $1/(1+\overline{\text{pop\_dev}}+\overline{\text{area\_dev}})$, and an L^∞ form $1/(1+\max_i[0.5\,\text{pop\_dev}_i+0.5\,\text{area\_dev}_i])$. DualBalance beats the enacted plan on the first two; the enacted plan wins on the third, because DualBalance concentrates area imbalance in a single rural district. We use the weighted form as the headline because it cleanly implements "each district carries $1/N$ of people *and* $1/N$ of geography." Per-district deviations are reported alongside so readers can recompute against any preferred aggregation.

The functional form satisfies four properties we wanted from the outset. (i) *Scale invariance*: the per-district inputs are relative deviations $|x - x^*|/x^*$, so the score is invariant to the units of population and area. (ii) *Equal treatment of the two objectives*: the 0.5/0.5 weighting is the unique convex combination treating $\overline{\text{pop\_dev}}$ and $\overline{\text{area\_dev}}$ symmetrically. (iii) *Monotonicity*: the score strictly decreases as either deviation increases. (iv) *Boundedness in* $[0, 1]$: the reciprocal form maps zero error to 1 and unbounded error toward 0, giving a calibrated reading rather than a raw cost. These do not pin DBS down uniquely, but they constrain the design space enough that the remaining choice (arithmetic mean of deviations versus a max-norm or Lp aggregation) becomes a tradeoff between average-case fairness and worst-case fairness. We report all the per-district numbers needed to compute both.

Population density variance bounds the achievable $\overline{\text{area\_dev}}$ on any pop-balanced plan. Both DualBalance and the enacted plan land near $\overline{\text{area\_dev}} \approx 1.0$–$1.1$. On Minnesota's geometry (Twin Cities = 55 % of population in 3 % of land area) this is a hard floor. Improvements would require districts that span urban and rural, which is exactly what radial slicing does. The 8.7-point gap (103.9 vs. 112.6) is the operational measure of how much area balance DualBalance recovers given that floor.

4.3 Compactness is the trade

DualBalance trades compactness for area balance by design. This is not an incidental cost we apologize for: it is the mechanism. A district that holds $1/N$ of a high-variance density distribution must either span the full density range (and look elongated) or sit inside a single density regime (and produce an unbalanced area share). Hand-drawn maps typically choose the second; DualBalance chooses the first. The PP=0.047 worst-slice number is what choosing the first looks like on Minnesota's geometry. The enacted Minnesota plan's worst PP is 0.178, roughly four times higher, and its $\overline{\text{area\_dev}}$ is correspondingly 8.7 points worse.

Three observations qualify the trade.

Compactness is a metric, not a constitutional requirement. *Polsby-Popper* as a measure was published in 1991 [12]; it appears in no federal statute and binds no court. Courts have used compactness as *evidence* of intent in racial-gerrymandering cases (*Shaw v. Reno* and progeny), but the constitutional violation in those cases turns on the use of race in line-drawing, not on the resulting shape per se. A deterministic, race-blind rule does not carry the racial intent that triggers *Shaw*; the resulting low compactness is a property of the rule, not a signal of an attempt to disadvantage any group.

Compactness assumes a baseline of comparison. A district is "bizarrely shaped" relative to surrounding districts that are not. If all eight districts in a state are produced by the same geometric rule and all eight share a similar slice-like character, the comparative baseline shifts. The aesthetic objection to long thin shapes is partly historical: it picks out districts that depart from the surrounding norm of compact blobs. A scheme in which every district is a slice does not pick anything out.

The Voting Rights Act trade-off is real. *Allen v. Milligan* [22] reaffirmed Section 2's effects-based vote-dilution doctrine; *Alexander* [24] and *Callais* [25] narrowed its practical reach. Section 2 asks about effects on minority voters' opportunity to elect, not about intent. DualBalance is race-blind, but blindness to race does not satisfy Section 2 by itself. In jurisdictions where the *Gingles* [19] preconditions are met, DualBalance may produce a plan that fails Section 2 review even though no one drew lines to dilute. The legislative question (whether to accept reduced minority opportunity in exchange for a generator that cannot be partisan-tuned) is for legislatures and courts, not for the algorithm.

4.4 Relationship to prior deterministic methods

DualBalance is, to our knowledge, the only deterministic districting method whose objective is bivariate (population *and* area). Every other published deterministic method optimizes a single extensive criterion – usually population balance plus a geometric proxy (compactness or shortest-cut length) – and that single-axis design is empirically why none of them clears all four of the criteria we set out in the introduction: legal viability under *Karcher*, DBS competitiveness, partisan-fairness on the gerrymandered states, and minority-majority district creation where the geography supports it. We argue this point empirically for the two methods we benchmark on all 50 states (Cascade and BDistricting) and structurally for the remainder.

Cascade (Iowa-LSA-flavored, county aggregation). Cascade prioritizes county integrity lexicographically: aggregate VTDs to counties, capacitated first-fit counties to districts, split a county only when its population exceeds the per-district cap. The county-integrity priority makes the algorithm legitimate *in Iowa*, where no county is large enough to dominate a district, but on every state with a major metropolitan county it produces population deviations far above *Karcher*: 0.50 % on Wisconsin, 10.27 % on North Carolina, 24.58 % on Texas, 41.56 % on Massachusetts, 76.14 % on Minnesota (Table 1). Cascade is therefore *not legally viable* on the majority of states, regardless of how well it does on DBS or efficiency gap. Treating Cascade as a counterfactual 'national rule' would constitutionally disqualify the resulting maps under *Reynolds v. Sims*, let alone *Karcher*.

BDistricting [9], representing the capacitated-k-means / centroidal-power-diagram family [3, 7]. BDistricting minimizes population-weighted Euclidean distance from each unit to its district center subject to equal-population caps, with Lloyd-style iteration to convergence – an empirical realization of the centroidal-power-diagram objective formalized by Cohen-Addad, Klein, and Young [3] and independently studied by Levin and Friedler [7]. The objective is unidimensional: equal population plus geometric compactness around dispersed centers. In our six-state comparison BDistricting achieves the highest Polsby-Popper and Reock scores of any algorithm on every state but is uneven on population (*Karcher* on some states, marginal on others) and middle-of-the-road on DBS because area balance is not part of the objective. The general pattern: a deterministic generator that optimizes population + compactness with no third axis cannot reliably hit the dual-balance objective, because the radial mixing of dense and sparse territory that gives DualBalance its area balance is explicitly destroyed by Lloyd recentering, which pulls seeds toward their cells' centers of mass.

Shortest Splitline [14]. Smith's algorithm recursively bisects the state with the shortest population-balancing line. The objective is unidimensional in a different sense: at each recursive step it minimizes one number (line length) and otherwise lets the geometry fall where it will. The resulting districts are often visibly elongated, with boundaries that follow neither county lines nor visible geographic features. No U.S. jurisdiction has adopted Splitline; no public 50-state baseline exists; the algorithm is exemplary as a proof of deterministic feasibility but not as a deployment candidate. We include it here for completeness rather than for empirical comparison.

Cascade as our chosen empirical baseline. We include Cascade in the comparison rather than Splitline because Cascade is the deterministic alternative that has actually been deployed at scale (in Iowa since 1980) and is the most-cited model of impartial districting in the popular literature. The empirical contribution of the comparison is to show that the Iowa template is geographically contingent: on the 49 other states, Cascade fails the legal-viability test catastrophically. The conceptual contribution is to clarify that being deterministic and impartial is necessary but not sufficient. The objective the algorithm pursues matters, and the DualBalance objective – equal population *and* equal area – is both expressible deterministically and broad enough to admit legally-viable plans on every state we have tested.

The forensic critique in §4.7 applies equally to all of these methods, and to any future deterministic generator built on similar principles: the gerrymandering metrics were built to investigate human line-drawers, and on any deterministic plan their inferential power evaporates.

4.5 Limitations

VTD data availability. The Census Bureau's TIGER 2020PL redistricting file includes VTD boundaries only for states that submitted them. At least three states — California, Hawaii, and Oregon — did not participate in the VTD program for 2020 and therefore have no `vtd20` shapefile in TIGER; they are excluded from all cross-state tables and figures here. Future work can substitute block-group or block boundaries, which the Census does publish for all states, at a cost of larger input size and longer optimizer runtime.

Peninsula and polycentric geometries. DualBalance's optimizer can fail to close the Karcher gap in states whose geometry defeats single-center radial seeding. Florida is the clearest case in our validation set: with 28 seats distributed across a long peninsula, the radial slices created around the population-weighted centroid produce districts that the VTD-scale optimizer cannot balance to the 0.05 % target in 10^5 passes, leaving $\text{pop\_dev}_{\text{max}} = 44.2\,\%$. The block-scale refinement pass (§2.8) is the next step for these hard-geometry states; results will be reported as they are available.

Expanding evidence base. Tables and figures in §?? will be updated as the full-state harness completes. Figures and tables note † for any state where VTD data were unavailable and results therefore could not be computed.

Worst-district area outlier. DualBalance concentrates the area imbalance in a single rural district (here, District 2 at 275 % over-target). This is geometric, not algorithmic: any pop-balanced partition of a state with Minnesota's density profile must give some rural district disproportionate area. DualBalance's $\text{area\_dev}_{\text{max}}$ of 275 % is worse than the enacted plan's 241 % even though the mean is lower, so DualBalance distributes the imbalance unevenly. Whether evenly distributed area imbalance is preferable to a single concentrated outlier is a separable normative question.

No multi-unit transportation step. The optional `--tighten-pop` pass is a greedy single-unit swap. A true 2D transportation LP bounding both population and area would directly minimize the DBS objective; we leave that to future work.

Computational cost. Pure DualBalance runs in under a second on Minnesota. The tightening pass takes ∼18 s for 80 swaps. We have not profiled California or Texas; worst-case scaling is $O(|U|^2 N)$ per swap due to the contiguity check.

The compactness floor. The worst slice's PP = 0.047 is well below the informal court-testimony threshold of 0.10. We argue in §4.3 that the *Shaw* [20] doctrine turns on intent rather than shape per se, but readers who reject that argument should regard the 0.047 figure as a hard ceiling on legal viability.

4.6 Future work

Broad-coverage validation (ongoing). The six-state initial validation (MN, IA, MA, TX, NC, WI) is being extended to the full set of multi-seat states for which TIGER 2020PL VTD boundaries are available (41 states at time of writing; CA, HI, OR excluded). Results will be incorporated into the published version of this paper. The most analytically important remaining states are Pennsylvania and Ohio (state-court partisan-gerrymandering jurisprudence) and the four states where DualBalance is still above the Karcher threshold at VTD scale (FL in particular, where the peninsula geometry makes single-center radial seeding hard to balance).

Multi-center radial seeding. For polycentric states, place seeds on circles around *each* of k population centers, with k chosen deterministically from the density distribution. Not yet implemented.

Direct 2D transportation step. Replace the DualBalance-plus-tighten two-stage pipeline with a single optimization that minimizes DBS under contiguity constraints. More expensive than the current pipeline; the mathematically clean answer to "which plan directly minimizes the DualBalance objective."

Splitline benchmark. The six-state benchmark (§**??**) covers DualBalance, Cascade, BDistricting, and the enacted plan. Adding Smith's Shortest Splitline [14] would give a fourth deterministic baseline; computing it from scratch on each state is straightforward but has not yet been done.

Scoring-harness extensions. The current harness reports population, area, Polsby-Popper, and Reock. Natural additions include partisan diagnostics (efficiency gap, mean-median, declination) on voter-registration data, county-split counts, and minority representation diagnostics under Section 2. These are reporting extensions and would not feed back into the generator (§2, "out-of-scope inputs").

4.7 Forensic metrics applied to a generative procedure

Two problems arise when forensic metrics are pressed into service as design objectives rather than as diagnostic tools. First, optimizing for a plan that "looks like what a neutral process would have produced" is a moving target: the counterfactual shifts with the ensemble distribution one chooses and the legal constraints one encodes, both of which carry discretionary content [1]. Second, fully content-neutral automated procedures can produce systematic partisan asymmetry as a geographic artifact. Chen and Rodden [2] showed that residential clustering causes impartial automated maps to disadvantage Democrats in many states, so a nonzero Efficiency Gap on such a map reflects geography rather than intent. Forensic metrics were designed to detect manipulation by a human line-drawer; absent a line-drawer, their inferential content evaporates [10].

Deterministic generation has interpretive consequences for the fairness-metrics literature. Most gerrymandering diagnostics carry two functions. They *describe* a property of the plan (a shape, a vote distribution, a count of split counties). And they support an *inference* about the line-drawer's motive (that the shape was chosen, that the votes were packed and cracked, that the splits followed a partisan or racial pattern). The two are nearly always entangled in practice because the literature was built around enacted plans, where intent is always at issue.

DualBalance separates them. The descriptive function survives intact: a PP of 0.09 still names a non-compact shape, with administrative and legitimacy consequences; an Efficiency Gap of +6.6 %

still names a real wasted-vote asymmetry, with representational consequences; zero majority-minority districts still names a partition under which no minority group reaches a 50 % VAP share, with Section 2 consequences. The inferential function does not. DualBalance is a pure function of geometry, population, and seat count, so the chain from observed property back to motive has no actor to terminate on.

Three implications, narrowed accordingly.

***Shaw*-style shape inference loses purchase, but compactness still matters administratively.** *Shaw v. Reno* [20] uses bizarre shape as evidence that race predominated in line-drawing. The inference chain runs shape → unusual process → illegitimate criterion → race. On DualBalance the first link is broken: every district produced by the rule has the same shape character, so unusual shape no longer implies unusual process. *Shaw* does not reach a race-blind algorithm. But compactness still has a non-inferential role. Districts that are hard to identify with on a map, hard to administer, or hard to campaign in carry costs independent of who drew them. DualBalance's PP=0.047 worst slice is a real cost. The argument here is not that compactness ceases to matter, only that it ceases to be evidence of motive.

Partisan metrics describe effects, not intent. *Rucho v. Common Cause* [21] foreclosed federal partisan-gerrymandering review for lack of a manageable standard for distinguishing acceptable from unacceptable partisan motivation. The question is moot on DualBalance (there is no motivation), but the partisan *effect* the metrics measure is not. An Efficiency Gap of +6.6 % on a DualBalance map still corresponds to wasted-vote asymmetry that affects representation; state-court tests that interpret EG as evidence of effect, rather than as evidence of intent, still bind. The rhetorical move from "the map shows EG +6.6 %" to "someone built that EG +6.6 %" becomes unavailable on DualBalance, but the metric continues to measure what it measured.

Ensemble outlier tests lose their baseline. The Duke and MGGG ensembles [4, 5] compare an enacted plan to a sample of legally compliant alternatives a neutral line-drawer might plausibly have produced. The sample stands in for the absent neutral line-drawer; the enacted plan being an outlier suggests the actual line-drawer differed from the neutral one. DualBalance *is* the neutral line-drawer in a stronger sense than the ensemble's sample mean. Asking whether the DualBalance plan is an outlier against its own ensemble may not be meaningful: the ensemble collapses to one point. Ensembles can still play a comparison role across *different* algorithmic generators (e.g. DualBalance vs. BDistricting vs. splitline), but the question is no longer "is this map suspiciously different from neutral?"

Metrics remain informative as descriptive summaries of plan effects. A DualBalance plan with $\mathrm{PP}_{\mathrm{min}} = 0.047$, Efficiency Gap +6.6 %, and zero majority-minority districts has low compactness, an R-favoring wasted-vote asymmetry, and no MMD. Those facts carry administrative, representational, and Section 2 consequences. The inference from those facts to a person who chose them is what deterministic generation removes.

4.8 What this paper claims, and what it does not

The paper makes four claims.

First, that a knob-free, race-blind, partisan-blind, deterministic districting algorithm exists that produces output competitive with, and on a single empirically tested state superior to, the corresponding hand-drawn enacted plan on a prespecified population-and-area objective. The algorithm's output is byte-reproducible and updates only with the ten-year census.

Second, that the algorithm's procedural neutrality is symmetric. The rule cannot be used to harm any group, party, or incumbent, and cannot be used to help any of them. The same property that forecloses a partisan gerrymander forecloses a race-conscious remedy. The symmetry is the algorithm's defining property; whether it is morally preferable to the status quo of partisan and remedial discretion is a question for legislatures, courts, and voters.

Third, that the post-*Callais* legal landscape gives a generator of this kind a narrower constitutional exposure than any race-conscious or partisan-conscious alternative. Federal partisan review is foreclosed under *Rucho*; federal racial-gerrymander risk is contracting under *Alexander* and *Callais*; state-court remedies for partisan claims depend on the politics of each state's supreme court. A rule whose inputs do not contain race or party is, in the strict sense, not subject to either body of doctrine.

Fourth, the conventional gerrymandering metrics (compactness scores, Efficiency Gap, mean-median, ensemble outliers, majority-minority counts) play two roles on enacted plans, only one of which transfers to DualBalance. As descriptions of plan effects (shape, wasted-vote asymmetry, minority opportunity) they remain valid and are reported alongside the DualBalance Score. As evidence of line-drawer intent they do not transfer. The intent-attribution use of these metrics, not their measurement use, is what the generative case breaks. This is the most contested framing claim in the paper.

The paper does not claim that DBS is universally the right objective, that DualBalance dominates enacted plans on all states, that compactness can be ignored, that effects-based fairness analyses (including Section 2) become irrelevant, or that the metric toolkit developed over the past thirty years is wrong. The narrow claim is that the intent reading of those metrics presupposes a line-drawer who is not present in this construction.

Whether deterministic generation is preferable to discretionary redistricting is a political and constitutional question; the algorithm itself cannot answer it.

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